

Pre-Course Diagnostic: Answer Key

How to use this key

Check your answers below. For each section, if you missed one question, just glance at the fix. If you missed most or all of a section, that's your signal to spend a little focused time there before Unit 1 — use the “What to review” guide at the very end to know exactly where to look.

Section 1: Order of Operations

1. $3 + 4 \times 2^2 = 3 + 4 \times 4 = 3 + 16 = 19$
2. $(5 - 2) \times (7 - 4) = 3 \times 3 = 9$
3. $12 \div 3 \times 2 = 4 \times 2 = 8$ (left to right, since division and multiplication have equal priority)

Section 2: Working with Fractions

4. $\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$
5. $\frac{2}{5} \times \frac{3}{4} = \frac{6}{20} = \frac{3}{10}$
6. $\frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = \frac{3}{2}$
7. $\frac{6}{8} = \frac{3}{4}$

Section 3: Negative Numbers

8. $-5 + 8 = 3$
9. $-3 \times (-4) = 12$
10. $7 - (-2) = 7 + 2 = 9$

Section 4: Exponents

11. $2^3 = 8$
12. $(-3)^2 = 9$ (the negative gets squared too)
13. $x^2 \cdot x^3 = x^5$ (add the exponents)
14. $5^0 = 1$ (anything nonzero to the power 0 is 1)

Section 5: Expanding and Distributing

15. $3(x + 4) = 3x + 12$
16. $(x + 2)^2 = x^2 + 4x + 4$
17. $(x + h)^2 = x^2 + 2xh + h^2$

18. $(a+3)(a-3) = a^2 - 9$

Section 6: Factoring

19. $x^2 + 5x + 6 = (x+2)(x+3)$

20. $x^2 - 9 = (x-3)(x+3)$

21. $4x + 8 = 4(x+2)$

Section 7: Function Notation and Evaluation

22. $f(3) = 3^2 + 1 = 9 + 1 = 10$

23. $f(-2) = 2(-2) - 5 = -4 - 5 = -9$

24. $f(a+h) = (a+h)^2 + 3(a+h) = a^2 + 2ah + h^2 + 3a + 3h$

25. $f(a+h) - f(a) = (a^2 + 2ah + h^2) - a^2 = 2ah + h^2$

Section 8: Simplifying Algebraic Fractions

26. $\frac{2xh + h^2}{h} = \frac{h(2x + h)}{h} = 2x + h$

27. $\frac{1}{x} - \frac{1}{2} = \frac{2}{2x} - \frac{x}{2x} = \frac{2-x}{2x}$

Section 9: Radicals (Square Roots)

28. $\sqrt{16} = 4$

29. $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$

30. $\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

31. $\sqrt{9} + \sqrt{4} = 3 + 2 = 5$

Section 10: Solving Linear Equations

32. $2x + 5 = 13 \Rightarrow 2x = 8 \Rightarrow x = 4$

33. $3(x-1) = 12 \Rightarrow 3x - 3 = 12 \Rightarrow 3x = 15 \Rightarrow x = 5$

Section 11: Slope and Lines

34. $\text{slope} = \frac{11-2}{4-1} = \frac{9}{3} = 3$

35. $y = 2x + 5$

Section 12: The Coordinate Plane

36. Quadrant II (negative x , positive y)

What to review, by section

Not all of these carry equal weight. Some skills are things the course genuinely assumes you already have; others get re-taught early on, so a shaky score there isn't a big red flag.

High priority — brush up before Unit 1 if you missed 2+ in these sections. These are assumed background and won't be retaught:

Section	Topic	Where it shows up
1	Order of operations	Everywhere, constantly
2	Fractions	Everywhere, constantly
3	Negative numbers	Everywhere, constantly
6	Factoring	Units 2, 6, 9, and most of Chapter 4
10	Solving linear equations	Everywhere
11	Slope and lines	Unit 1 itself, and tangent lines throughout the course

Lower priority — don't stress if these felt rusty. The course actually re-teaches these skills, step by step, right at the start:

Section	Topic	Where it's taught (or re-taught)
4	Exponents	Reviewed naturally in Units 1, 3, and 9
5	Expanding binomials like $(a + h)^2$	This is the core skill of Unit 1 itself — you'll get heavy practice there
7	Function notation, $f(a + h)$	Also central to Unit 1 — expect a lot of repetition
8	Simplifying algebraic fractions	Practiced directly in Units 1 and 2
9	Radicals and rationalizing	Explicitly taught as a technique in Units 1 and 2 (the “conjugate trick”)
12	Coordinate plane basics	Used lightly throughout; picks up naturally

Bottom line: if your weak spots are concentrated in Sections 1, 2, 3, 6, 10, or 11, take a short detour before starting — a quick review video or practice set on fractions, factoring, or solving equations will pay off immediately. If your weak spots are in Sections 4, 5, 7, 8, or 9, don't worry — just start Unit 1 as planned, since the course builds those skills back up from scratch anyway.